

Automated Requirement Classification From Technical Specifications

EECE5644: Machine Learning and Pattern Recognition

Final Project

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Problem Statement



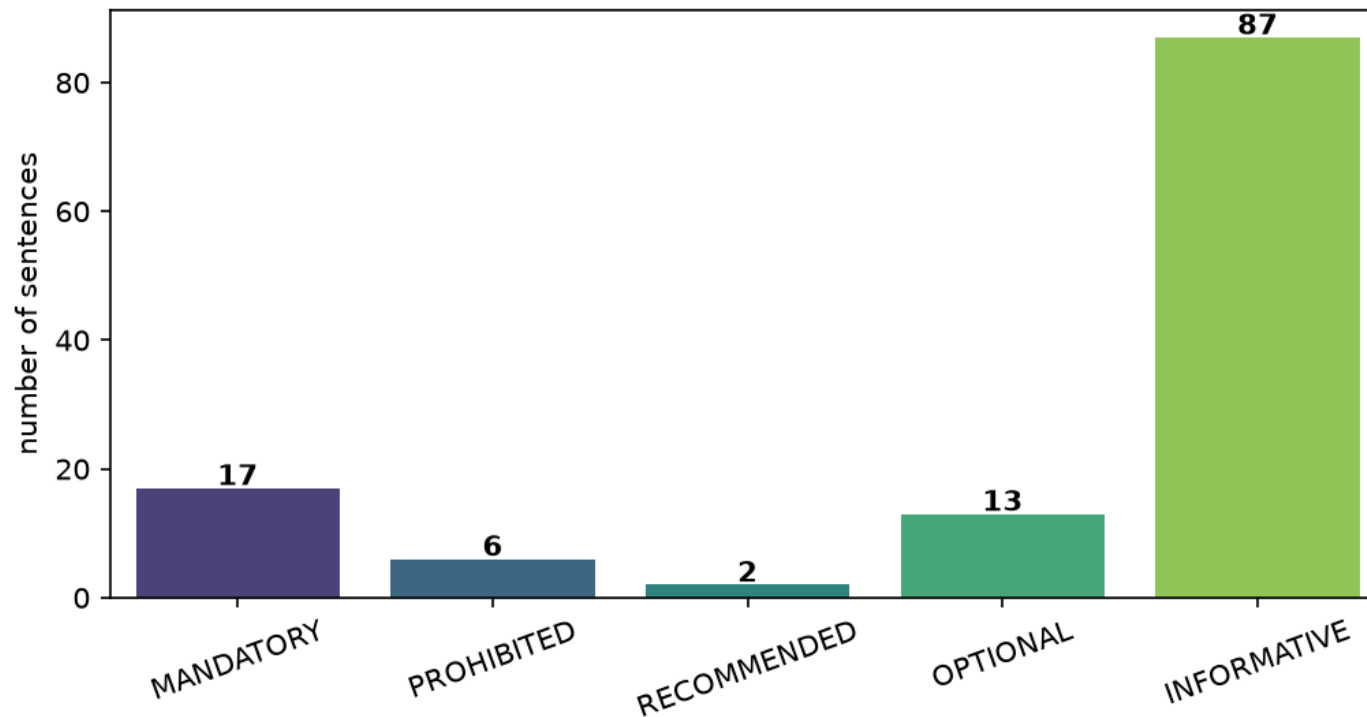
- Technical specs (NVMe, PCIe, CXL) encode a different obligation level in almost every sentence.
 - MANDATORY ("shall"), PROHIBITED ("shall not"), RECOMMENDED ("should"), OPTIONAL ("may"), INFORMATIVE (prose).
- Engineers must separate and type requirements by hand before writing validation tests slow and error-prone.
- Goal: automatically (1) extract requirement sentences and (2) classify each by normative modality.
- This is the requirement-understanding stage of a larger document-to-code agent.

- End-to-end, reproducible pipeline:
 - parse → segment → filter → weakly label → split → train → evaluate.
- Stage 1: Document understanding: Docling → structured Markdown/JSON with section headers, page provenance, and block type.
- Stage 2: Requirement classification: two model families trained and evaluated on identical stratified splits and metrics.
- One command regenerates every metric and figure; nothing between the raw PDF and the reported numbers is hand-edited.
- Modular design mirrors a larger “document-to-code” agent; this project delivers its first two stages.

Dataset & Weak Labelling



Class distribution (modality labels)



- Source: NVM Express Base Spec, Rev 2.3.
- Parsed with Docling sentence records with page & section.
- 125 labeled sentences.
- No gold labels weak supervision from modal keywords.
- Realistic, heavy imbalance: INFORMATIVE dominates; RECOMMENDED/PROHIBITED rare.
- Stratified train / val / test split.

Weak Supervision Labels Without Gold Data

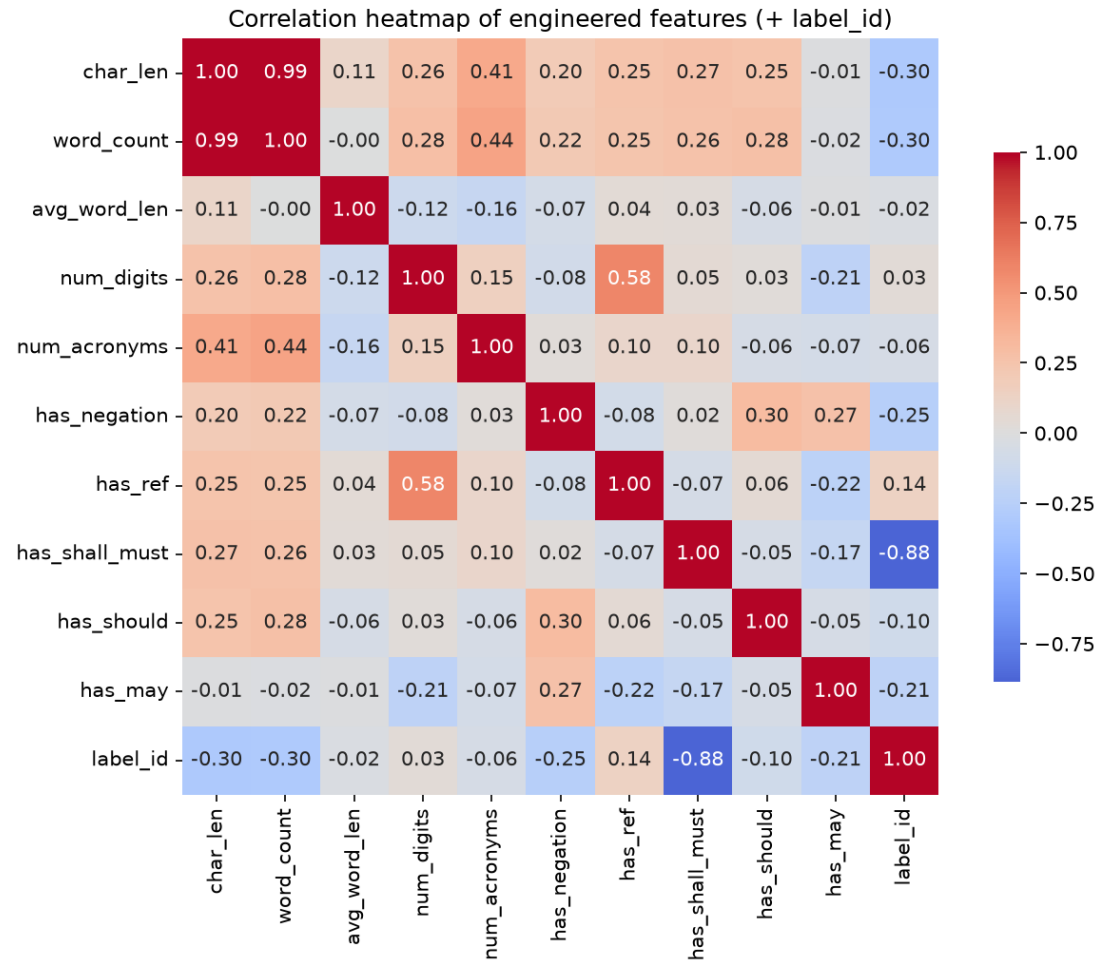


- Specs carry no gold labels → labels come from the document's own modal language (config.MODALITY_RULES).
- Curated, longest-match-first lexicon with explicit precedence:
 - “shall not” / “may not” matched before “shall” / “may”; word-boundary regex so “may” never fires inside “maybe.”
- A standard weak-supervision approach in requirements-engineering NLP scalable, transparent, annotation-free.
- Trade-off: some irreducible label noise “may” as permission vs. possibility can be mislabeled.

- Keep only Docling body prose (text / paragraph / list_item); drop tables, figures, headers.
- Regex sentence segmentation that protects spec abbreviations (e.g., i.e., Fig., No., vol.).
- Drop fragments < 15 chars or with no letters (removes numbers & cross-ref debris).
- Remove exact-text duplicates before labeling.
- Word-boundary label matching: "shall not" before "shall"; "may" doesn't fire inside "maybe".

- Classical models TF-IDF: sublinear TF, 1–2 grams (captures "shall not", "may not"), accent stripping.
- DistilBERT: WordPiece tokenization (max length 128), fine-tuned contextual embeddings.
- Interpretable engineered features for EDA:
 - length, avg word length, digit & acronym counts, negation / cross-ref / modal-cue flags.
- Class imbalance handled with class weights (both model families).

Exploratory Data Analysis

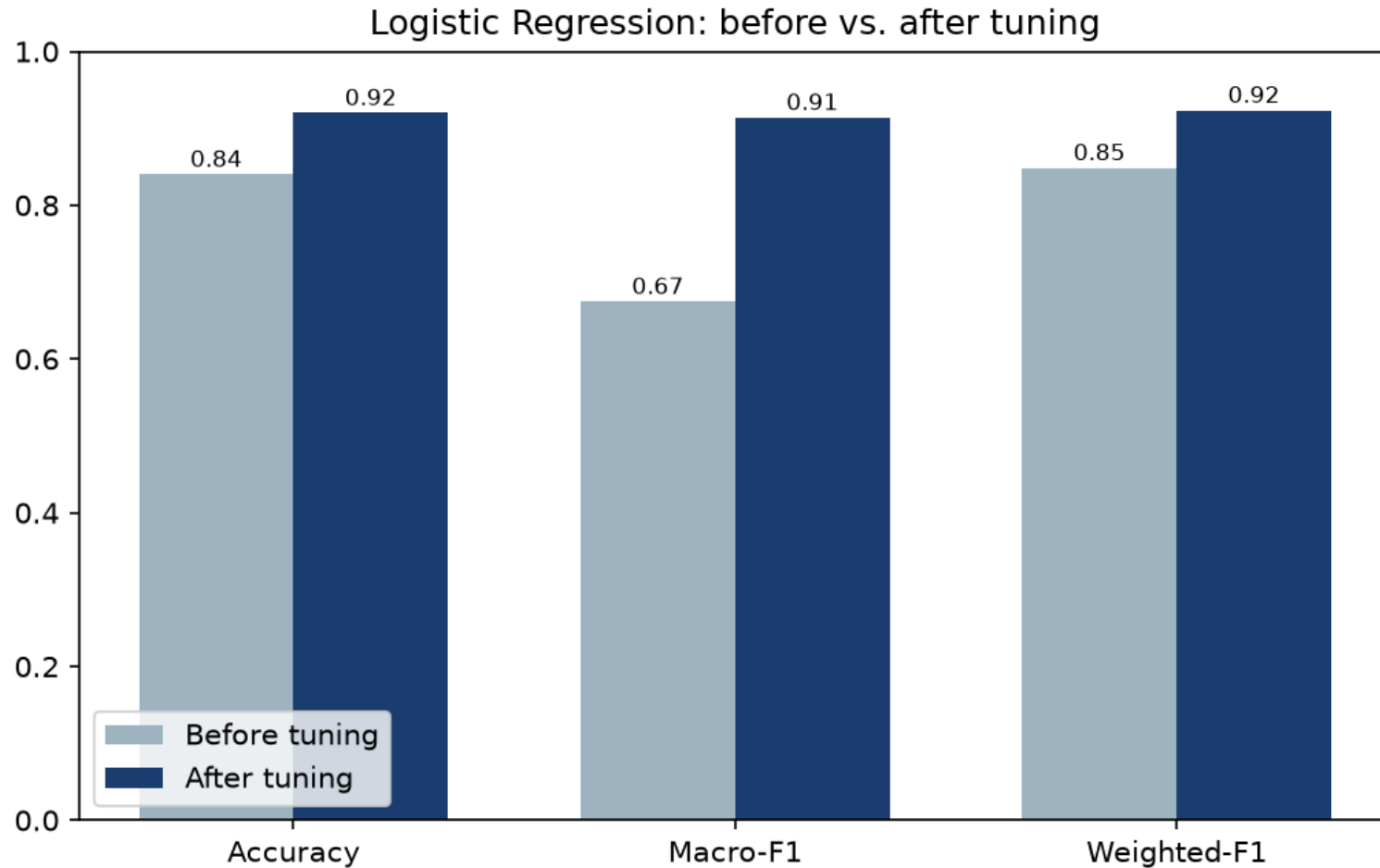


- No single surface feature strongly correlates with the label.
- Modal-cue flags carry the most signal.
- Sentence length distributions overlap across classes.
- The class boundary lives in word choice & context, not simple statistics.



- Multinomial Naive Bayes generative MAP baseline over TF-IDF counts.
- Logistic Regression (TF-IDF) discriminative, balanced, interpretable coefficients.
- Linear SVM (TF-IDF) max-margin in high-dimensional sparse space.
- DistilBERT 6-layer distilled BERT + linear head, class-weighted cross-entropy loss.
- Classical model selected by stratified k-fold CV (macro-F1); DistilBERT by val macro-F1.

Hyperparameter Tuning



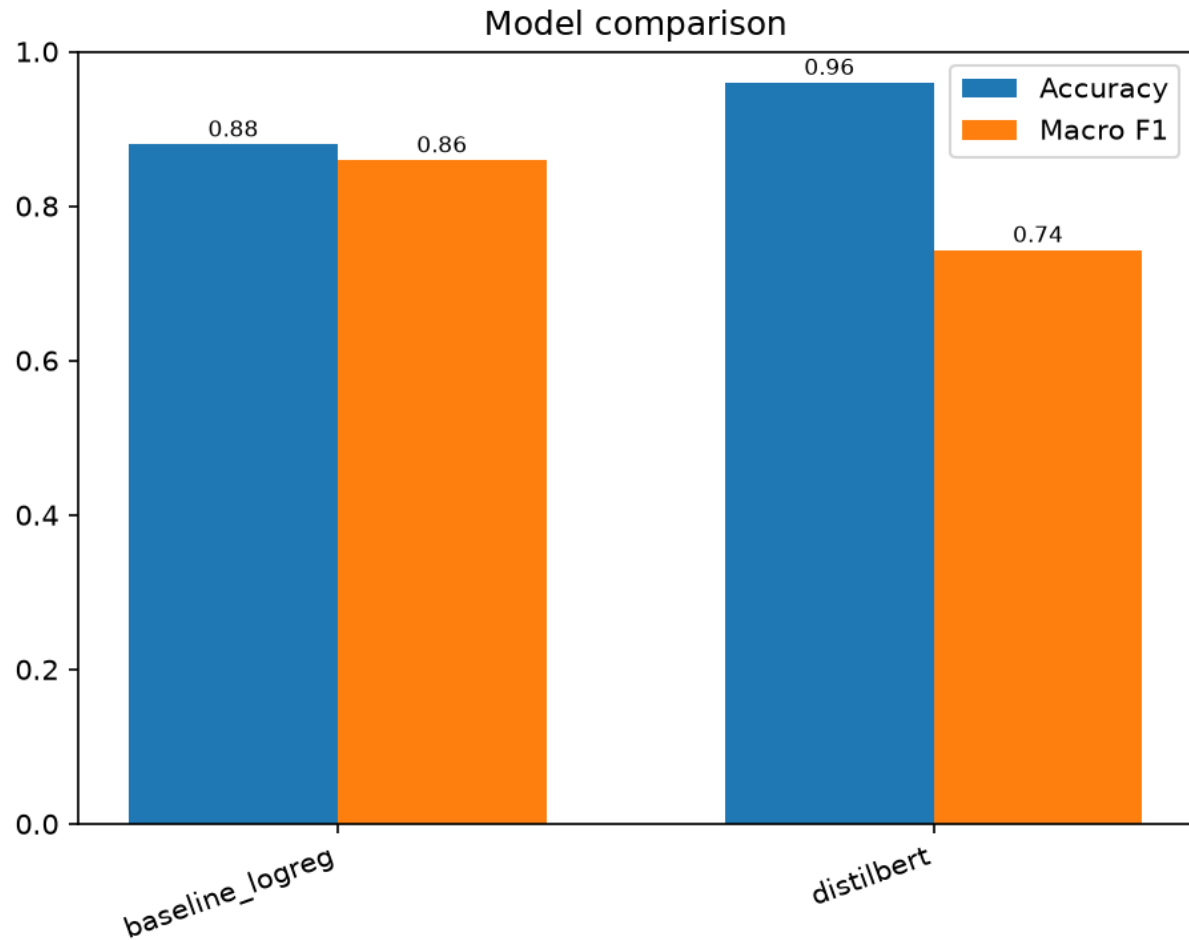
- Grid search: 24 combos $\times$ 2-fold CV (macro-F1).
- Tuned `ngram_range`, `min_df`, `sublinear_tf`, `C`.
- Best: bigrams, `min_df`=2, sublinear TF, `C`=1.0.
- Test macro-F1 0.67 - 0.91 (+0.24).
- Biggest gains: bigrams + sublinear TF scaling.

Evaluation Protocol & Metric Choice



- Stratified 80 / val / 20 split (seed 5644); test set ($n = 25$) is never seen during training or model selection.
- Classical models selected by stratified k-fold CV on macro-F1; DistilBERT by best-validation-checkpoint macro-F1.
- Headline metric = macro-F1: a majority-only classifier already scores ≈ 0.70 accuracy, so accuracy alone is misleading.
- Macro-F1 averages per-class F1 equally $\rightarrow$ it rewards getting the rare, high-value modalities right.
- ROC / PR (one-vs-rest): AUC ≈ 1.0 for well-represented classes; PROHIBITED is low purely from 1-sample support.

Results Model Comparison



- DistilBERT: accuracy 0.96, weighted-F1 0.94.
- Tuned LogReg: best macro-F1 0.91.
- Majority baseline accuracy ≈ 0.70 .
- Macro-F1 is the headline metric under imbalance.
- Linear model rivals the Transformer on this task.

Why a Linear Model Rivals the Transformer



- DistilBERT wins accuracy (0.96) & weighted-F1 (0.94); tuned LogReg wins macro-F1 (0.91) the imbalance metric.
- The signal is largely lexical: modal cues (“shall”, “may not”) are near-linearly separable in TF-IDF space.
- DistilBERT’s lower macro-F1 (0.74) comes almost entirely from the 1-sample PROHIBITED class variance, not capacity.
- Masking ablation: both stay above the 0.70 baseline with the trigger hidden → real context use, not keyword lookup.
- Practical read: prefer the interpretable linear model here; revisit the Transformer as rare-class data grows.

Results Error Analysis



Confusion matrix -- distilbert

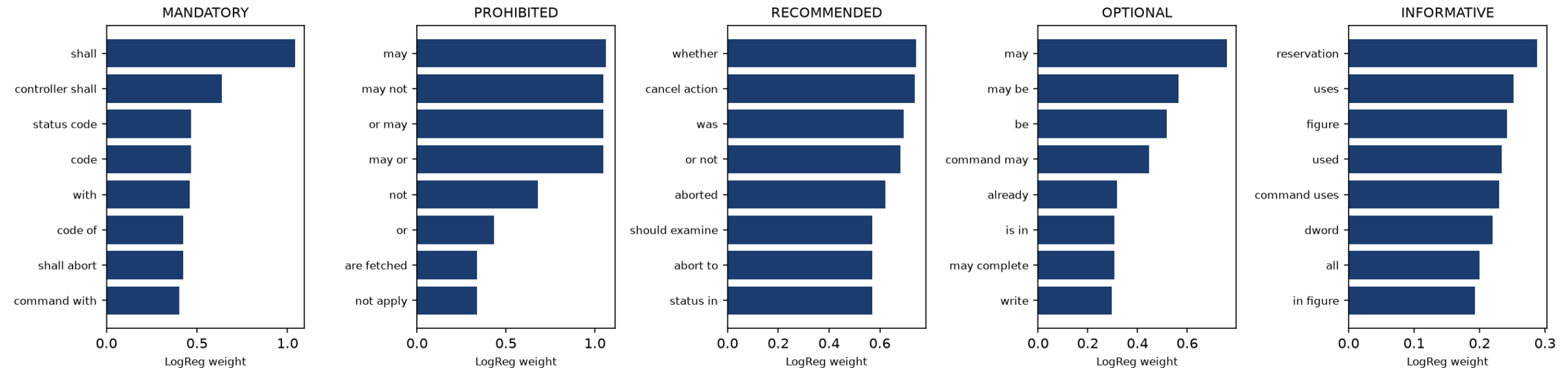
True label \ Predicted label	MANDATORY	PROHIBITED	RECOMMENDED	OPTIONAL	INFORMATIVE
MANDATORY	3	0	0	0	0
PROHIBITED	0	0	0	0	1
RECOMMENDED	0	0	0	0	0
OPTIONAL	0	0	0	3	0
INFORMATIVE	0	0	0	0	18

- Perfect on MANDATORY & OPTIONAL (P=R=1.00).
- INFORMATIVE F1 $\approx$ 0.97.
- Errors sit in PROHIBITED (1 test sample) & RECOMMENDED (0).
- Data-scarcity artifact, not a systematic weakness.
- Critical error (dropping a real requirement) is minimized: MANDATORY recall = 1.00.

Feature Importance & Learning Curve



Top TF-IDF features per class (Logistic Regression weights)



- Top terms per class are intuitive:
- MANDATORY: "shall", "controller shall".
- OPTIONAL/PROHIBITED: "may", "may not".
- Generic nouns & "figure"/"section" $\approx$ zero weight.
- Learning curve still rising - more data would help.

Do the Models Learn Meaning, or Just Keywords



- Ablation: blank the modal trigger word and re-evaluate.
- Logistic Regression accuracy: 0.88 (full) - 0.80 (masked).
- DistilBERT accuracy: 0.96 (full) - 0.80 (masked).
- Both stay above the 0.70 majority baseline with the keyword hidden.
- The models use sentence context, not a keyword lookup.

Challenges & Lessons Learned



- Extreme class imbalance with only 0–2 samples for the rarest classes on the test set.
- Weak labels are noisy "may" as permission vs. possibility can be mislabeled.
- Tiny test set makes macro-F1 high-variance.
- Lesson: a well-tuned linear model can match a Transformer on small, imbalanced text.
- Lesson: data volume for rare classes not model capacity is the real bottleneck.



- Use: feed a document-to-code agent typed requirements → test plans, traceability, coverage.
- Beneficiaries: firmware/hardware validation, compliance, and technical-writing teams.
- Future: expand corpus & add more standards (learning curve still climbing).
- Future: build a human-verified gold test set to remove weak-label noise.
- Future: add downstream requirement - test-case generation; confidence calibration for human-in-the-loop.

Thank You

https://github.com/ajaynathch/requirement_doc_parser

